B.Sc. Mathematics — Semester I

Paper-II: Differential Equations

Complete Exam Kit — Backup Master Copy

Structured copy of the supplied syllabus, complete question bank, and supplied examination papers.

B.Sc. Mathematics — Semester I

Paper-II: Differential Equations

Complete Exam Kit — Backup Master Copy

COURSE SYLLABUS

UNIT I: Differential Equations of First Order and First Degree

• Exact Differential Equations: Definition, necessary and sufficient condition for exactness M(x,y)dx + N(x,y)dy = 0.

• Integrating Factors: Methods of finding integrating factors for non-exact equations:

• By inspection.

• 1/(Mx + Ny) for homogeneous equations.

• 1/(Mx - Ny) for equations of the form f_1(xy)y dx + f_2(xy)x dy = 0.

• exp(∫ f(x)dx) when (1/N)(∂M/∂y - ∂N/∂x) = f(x).

• exp(∫ g(y)dy) when (1/M)(∂N/∂x - ∂M/∂y) = g(y).

• Linear Differential Equations: First-order linear equations dy/dx + Py = Q.

• Bernoulli's Differential Equation: Reducible to linear form dy/dx + Py = Qy^n.

• Orthogonal Trajectories: Cartesian and Polar coordinates.

UNIT II: Equations of First Order Higher Degree & Applications

• Equations of First Order but Not First Degree:

• Equations solvable for p (where p = dy/dx).

• Equations solvable for y.

• Equations solvable for x.

• Clairaut's Equation: y = px + f(p) and equations reducible to Clairaut's form.

• Applications of First Order Differential Equations:

• Growth and Decay problems.

• Radioactivity and Carbon Dating.

• Newton's Law of Cooling.

• Rate of Compound Interest.

UNIT III: Higher Order Linear Differential Equations

• Linear Differential Equations with Constant Coefficients: Complementary Function (CF) and Particular Integral (PI).

• Methods for Finding Particular Integrals:

• P.I. = 1/f(D^2) sin ax or 1/f(D^2) cos ax.

• Method of Undetermined Coefficients.

• Method of Variation of Parameters.

UNIT IV: Special Equations, Simultaneous & Total Differential Equations

• Cauchy-Euler Equations: x^n d^n y/dx^n + a_1 x^(n-1) d^(n-1)y/dx^(n-1) + ... + a_n y = X.

• Legendre's Linear Equations: (ax + b)^n d^n y/dx^n + ... + a_n y = X.

• Simultaneous Linear Differential Equations with constant coefficients.

• Total Differential Equations:

• Condition for integrability of P dx + Q dy + R dz = 0.

• Methods of solution: Inspection, Variable separation, Homogeneous equations, Auxiliary equations (dx/P = dy/Q = dz/R).

UNIT I

SECTION A — 5★ MOST IMPORTANT QUESTIONS

1. Solve y - x dy/dx = a (y^2 + dy/dx)
2. Solve 3e^x tan y dx + (1 - e^x) sec^2 y dy = 0
3. Solve dy/dx = sin(x + y) + cos(x + y)
4. Solve sqrt(1 + x^2) dx + sqrt(1 + y^2) dy = 0
5. Solve (x + y)^2 dy/dx = a^2
6. Solve (x^2 - y^2) dx + 2xy dy = 0
7. Solve dy/dx = (x + 7y + 2)/(3x + 5y + 6)
8. Solve (1 + e^(x/y)) dx + e^(x/y)(1 - x/y) dy = 0
9. Solve (1 + x) dy/dx - xy = 1 - x
10. Solve dy/dx + (ax + hy + g)/(hx + by + f) = 0
11. Solve (1 + xy)y dx + (1 - xy)x dy = 0

SECTION A — 3★ IMPORTANT QUESTIONS

1. Solve x sqrt(1 + y^2) dx + y sqrt(1 + x^2) dy = 0
2. Solve (1 - x) dy + (1 - y) dx = 0
3. Solve dy/dx = (x + 3y + 5)/(2x + 6y + 7)
4. Solve dy/dx - csc 2xy = 1/2 sec^2 x
5. Solve (sin x cos y + e^(2x)) dx + (cos x sin y + tan y) dy = 0

SECTION B — 5★ MOST IMPORTANT QUESTIONS

1. Solve y dx + (x - y^3) dy = 0
2. Solve y(x + y) dx - x^2 dy = 0
3. Solve (2x + y - 3) dy = (9x + 2y - 3) dx
4. Solve (1 + y^2) dx = (tan^(-1) y - x) dy
5. Solve dy/dx + x sin 2y = x^3 cos^2 y
6. Solve (x^3 - 2y^2) dx + 2xy dy = 0
7. Solve x dx + y dy = a^2 ((x dy - y dx)/(x^2 + y^2))

SECTION B — 3★ IMPORTANT QUESTIONS

1. Solve (1 - x^2) dy/dx + 2xy = x sqrt(1 - x^2)
2. Solve (1 - x^2) dy/dx + xy = xy^2
3. Solve (3x^2 y^4 + 2xy) dx + (2x^3 y^3 - x^2) dy = 0
4. Solve (y^2 + 2x^2 y) dx + (2x^3 - xy) dy = 0
5. Solve sec^2 y dy/dx + 2x tan y = x^3

UNIT II

SECTION A — 5★ MOST IMPORTANT QUESTIONS

1. Solve x^2 p^2 + 3xyp + 2y^2 = 0 where p = dy/dx
2. Solve (p + y + x)(xp + y + x)(p + 2x) = 0
3. Solve p^2 - 7p + 12 = 0
4. Solve xp^2 - 2yp + x = 0 where p = dy/dx
5. Solve y + px = x^4 p^2
6. Solve p = log(px - y)
7. Solve (y - px)(p - 1) = p
8. Write in detail about differential equations of first order but not of first degree.
9. Find the orthogonal trajectories of y^2 = 4ax.

SECTION A — 3★ IMPORTANT QUESTIONS

1. Find the singular solution of y = xp - p^2 where p = dy/dx
2. Solve y^2 log y = xyp + p^2
3. Find the general solution of y = xp + a/p
4. If ₹10,000 is invested at 6% per annum, find what amount has accumulated after 6 years if interest is compounded continuously.
5. Find the orthogonal trajectories of x^2 + y^2 = a^2.

SECTION B — 5★ MOST IMPORTANT QUESTIONS

1. Solve p^2 + 2py cot x = y^2
2. Solve y = yp^2 + 2px
3. Solve y = 3px + 6p^2 y^2
4. Reduce xyp^2 - (x^2 + y^2 - 1)p + xy = 0 to Clairaut's form and find its singular solution.
5. Write a short note on radioactivity and carbon dating.
6. How long does it take for a given amount of money to double at 6% per annum compounded (a) annually (b) continuously?
7. Find the orthogonal trajectory of r = c_1(1 - sin θ).
8. Find the orthogonal trajectories of x^(1/3) + y^(1/3) = c_1.
9. Find the family orthogonal to the family y = c e^(-x) of exponential curves. Determine the member of each family passing through (0, 4).

SECTION B — 3★ IMPORTANT QUESTIONS

1. Solve y - 2px = x^2 p^4
2. It is found that 0.5 percent of radium disappears in 12 years. (a) What percentage will disappear in 1000 years? (b) What is the half-life of radium?
3. Solve (p - xy)(p - x^2)(p - y^2) = 0
4. Solve x^2 p^2 + yp(2x + y) + y^2 = 0 by reducing it to Clairaut's form by using the substitution y = u and xy = v.
5. Bacteria in a certain culture increase at a rate proportional to the number present. If the number doubles in one hour, how long does it take for the number to triple?
6. Bacteria in a certain culture increases at a rate proportional to the number present. If the number N increases from 1000 to 2000 in 1 hour, how many are present at the end of 1.5 hours?
7. Write a short note on compound interest.
8. Find the orthogonal trajectories of x^2 + y^2 = cx.

UNIT III

SECTION A — 5★ MOST IMPORTANT QUESTIONS

1. Solve d^2y/dx^2 - 2dy/dx - 15y = 0
2. Solve d^2y/dx^2 + 16y = 0
3. Solve d^3y/dx^3 + y = 0
4. Solve d^2y/dx^2 - 3dy/dx + 2y = 3 sin x
5. Find a particular integral of (D^2 + 4D + 3)y = cos(3x)
6. Solve (D^2 - 4)y = x^2
7. Solve (D^2 + 2D + 1)y = 2x + x^2
8. Solve (D^2 - 2D + 5)y = e^(2x) sin x
9. Solve d^2y/dx^2 - 2dy/dx + y = x sin x
10. Solve (3D^2 + D - 14)y = 13e^(3x) where D = d/dx
11. Solve y'' + 3y' + 2y = 12e^x
12. Solve (D^2 - 3D + 2)y = 2x^2 + 3e^(2x) by using the method of undetermined coefficients.

SECTION A — 3★ IMPORTANT QUESTIONS

1. Solve d^3y/dx^3 - 6d^2y/dx^2 + 11dy/dx - 6y = 0
2. Solve d^3y/dx^3 + 6d^2y/dx^2 + 11dy/dx + 6y = 0
3. Solve d^2y/dx^2 + (a + b)dy/dx + ab y = 0
4. Solve (D^2 - 2mD + m^2)y = e^(mx) where D = d/dx
5. Find a particular integral of (D^2 - 3D + 2)y = γ sin(3x)
6. Solve (D^2 - 3D + 2)y = sin 2x
7. Solve d^2y/dx^2 = x^2 + 2x
8. Solve (D^2 + 4D + 4)y = 4x^2 + 6e^x by using method of undetermined coefficients.

SECTION B — 5★ MOST IMPORTANT QUESTIONS

1. Write about nth order homogeneous differential equations.
2. Solve (D^2 - 4D + 4)y = 8(x^2 + e^(2x) + sin 2x)
3. Solve (D^2 + 4D + 4)y = e^(-2x) + x^2
4. Solve (D^2 + 4)y = x cos x
5. Solve d^2y/dx^2 + 2dy/dx + y = x cos x
6. Solve (D^2 - 2D + 1)y = x e^x sin x
7. Solve (D^2 + 3D + 2)y = x e^(2x) + sin x by using the method of undetermined coefficients.
8. Solve (D^2 + 4D + 4)y = 3x e^(-2x) using undetermined coefficients.
9. Solve (D^2 + 9)y = cos 3x + sin 2x

SECTION B — 3★ IMPORTANT QUESTIONS

1. Write the solution of nth order differential equation of the type f(D)y = X.
2. Find the particular integral of (D^3 - D^2 - 6D)y = x^2 + 1.
3. Solve (D^3 - 1)y = x e^x
4. Solve (D^2 + 1)y = x^2 sin 2x
5. Solve (D^2 + 1)y = cos x + x e^(2x) + e^x sin x

UNIT IV

SECTION A — 5★ MOST IMPORTANT QUESTIONS

1. Explain the method of variation of parameters.
2. Given that y_1 = x is a solution of x^2 d^2y/dx^2 + x dy/dx - y = 0, x ≠ 0. Then find the general solution of x^2 d^2y/dx^2 + x dy/dx - y = x.
3. Solve x^2 y'' + y = 0, given y_1 = x as a solution.
4. Solve x^2 d^2y/dx^2 - x dy/dx + y = 2 log x.
5. Solve (9x^2 D^2 + 3xD + 1)y = 0.
6. Solve d^2y/dx^2 = x e^x.
7. Solve dx/(1 + x) = dy/(1 + y) = dz/z.
8. Solve dx/x = dy/(x + z) = dz/(-z).
9. Solve x dx + z dy + (y + 2z) dz = 0.
10. Solve (x^2 D^2 + 3xD + 1)y = 0.

SECTION A — 3★ IMPORTANT QUESTIONS

1. Solve 2x^2 y'' + 3xy' - y = 0 given that y_1 = 1/x is a solution.
2. Explain the Cauchy-Euler equation.
3. Explain Legendre's linear equation.
4. Solve x dx + z dy + (y + 2z) dz = 0.
5. Solve dx/(z^2 y) = dy/(z^2 x) = dz/(y^2 x).
6. Solve dx/(mz - ny) = dy/(nx - lz) = dz/(ly - mx).

SECTION B — 5★ MOST IMPORTANT QUESTIONS

1. Using the method of variation of parameters, solve y'' + 9y = csc(3x).
2. Solve (D^2 - 3D + 2)y = sin(e^(-x)) using the method of variation of parameters.
3. Using the method of variation of parameters, solve (D^2 - 1)y = 1/(1 + e^x).
4. Solve x^3 D^3 y + 3x^2 D^2 y + xDy + y = x + log x.
5. Solve x^3 y''' + 3x^2 y'' + xy' + 8y = 65 cos(log x).
6. Solve (1 + x)^2 d^2y/dx^2 + (1 + x)dy/dx + y = 2 sin[log(1 + x)].
7. Solve (y^2 + yz) dx + (z^2 + zx) dy + (y^2 - xy) dz = 0.
8. Solve dx/[x(y^2 - z^2)] = dy/[y(z^2 - x^2)] = dz/[z(x^2 - y^2)].
9. Solve dx/[x(y^2 - z^2)] = dy/[-y(z^2 + x^2)] = dz/[z(x^2 + y^2)].
10. Solve x^2 y'' - 6y = 5x^3 + 8x^2.

SECTION B — 3★ IMPORTANT QUESTIONS

1. Solve (D^2 + 4D + 4)y = 3x e^(-2x) by using the method of variation of parameters.
2. Solve y'' + 3y' + 2y = 12e^x, by using the method of variation of parameters.
3. Using the method of variation of parameters, solve the equation y'' + y = 4x sin x.
4. Solve x^2 D^2 y - xDy - 3y = x^2 log x.
5. Solve x^2 d^2y/dx^2 + x dy/dx - 4y = x^2 by using Cauchy-Euler equation.
6. Solve the differential equation d^2y/dx^2 = 2(y^3 + y) under the condition y = 0, dy/dx = 1, when x = 0.

B.A./B.Sc. (CBCS) I-Semester Examination, August 2026

FACULTY OF SCIENCE

Subject: Mathematics | Paper - I: Differential Equations

Code No: H-1205/BL | Time: 3 Hours | Max. Marks: 80

PART - A

(8 × 4 = 32 Marks — Answer any Eight questions)

1. Solve dy/dx = x(2log x + 1)/(sin y + y cos y)
2. Solve dy/dx = (4x + y + 1)^2
3. Solve (x^2 + y^2 + 2x)dx + 2y dy = 0
4. Solve p^2 - 7p + 12 = 0, where p = dy/dx
5. Solve y = 3x + log p, where p = dy/dx
6. How long does it take for a given amount of money to double at 6 per cent per annum compounded annually?
7. Solve (D^3 + 1)y = 0, where D = d/dx
8. Solve (D^2 - 2D + 5)y = e^(-x), where D = d/dx
9. Solve (D^3 - D^2 - D + 1)y = 0, where D = d/dx
10. Solve dx/(x^2 y) = dy/(z^2 y) = dz/(y^2 x)
11. Solve dx/(mz - ny) = dy/(nx - lz) = dz/(ly - mx)
12. Solve yz dx - 2xz dy + (xy - zy^3)dz = 0

PART - B

(4 × 12 = 48 Marks — Answer all questions)

13. a) Solve (1 + e^(x/y))dx + e^(x/y)(1 - x/y)dy = 0

OR

b) Solve (xy sin xy + cos xy)y dx + (xy sin xy - cos xy)x dy = 0

14. a) It is found that 0.5 per cent of radium disappears in 12 years. What percentage will disappear in 1000 years?

OR

b) Find the orthogonal trajectories of the family of parabolas opening in the y-direction with the vertex at (1,2).

15. a) Solve (D^2 + 4D - 12)y = (x - 1)e^(2x)

OR

b) Solve (D^2 - 2D + 1)y = x e^x sin x

16. a) Solve d^2y/dx^2 + 3dy/dx + 2y = 12e^x by the method of undetermined coefficients.

OR

b) Solve d^2y/dx^2 - 2dy/dx + y = e^x log x by the method of variation of parameters.

B.A./B.Sc. (CBCS) I-Semester Examination, January 2026

FACULTY OF SCIENCE

Subject: Mathematics | Paper - I: Differential Equations

Code No: H-1206 | Time: 3 Hours | Max. Marks: 80

PART - A

(8 × 4 = 32 Marks — Answer any Eight questions)

1. Solve 3e^x tan y + (1 - e^x)sec^2 y dy/dx = 0
2. Solve cos^2 x dy/dx + y cos x = sin x
3. Solve y(1 - xy)dx - x(1 + xy)dy = 0
4. Solve p^2 + 2py cot x = y^2, where p = dy/dx
5. Solve (y - px)(p - 1) = p, where p = dy/dx
6. Solve x^2 + p^2 x = yp, where p = dy/dx
7. Solve (D^2 + (a + b)D + ab)y = 0, where D = d/dx

8. Solve (D^3 - 3D + 2)y = 0, where D = d/dx

9. Solve d^2y/dx^2 - y = sin x

10. Solve (y + z)dx + (z + x)dy + (x + y)dz = 0

11. Solve dx/x^2 = dy/y^2 = dz/(nxy)

12. Solve x^2 y'' - xy' + y = 0, given y_1 = x as a solution.

PART - B

(4 × 12 = 48 Marks — Answer all questions)

a) Solve (3x - y + 1)dx + (2y - x - 1)dy = 0

OR

b) Solve (1 - x^2)dy/dx + xy = xy^2

a) Solve x^2 p^2 + yp(2x + y) + y^2 = 0 by reducing it to Clairaut's form using the substitution y = u and xy = v.

OR

b) Bacteria in a certain culture increase at a rate proportional to the number present. If the number doubles in one hour, how long does it take for the number to triple?

a) Solve (D^2 - 2D + 5)y = 0 given that y = 0 and dy/dx = 4 when x = 0, where D = d/dx

OR

b) Solve (D^3 - D^2 - 6D)y = x^2 + 1, where D = d/dx

a) Solve x^2 d^2y/dx^2 + x dy/dx - 4y = x^2

OR

b) Solve d^2y/dx^2 = 2(y^3 + y) under the condition y = 0 and dy/dx = 1 when x = 0.

Osmania University — June 2025

Time: 3 Hours | Max. Marks: 80

PART-A

(8 × 4 = 32 Marks — Answer any eight questions)

1. Solve (x + y)^2 dy/dx = a^2

2. Solve (1 + xy)y dx + (1 - xy)x dy = 0

3. Solve x dx + z dy + (y + 2z) dz = 0

4. Solve p^2 + 2py cot x = y^2

5. Solve y^2 log y = xyp + p^2

6. If ₹10,000 is invested at 6 percent per annum, find what amount has accumulated after 6 years if interest is compounded continuously.

7. Solve d^3y/dx^3 + y = 0

8. Solve (D^2 - 3D + 2)y = sin 2x

9. Solve d^2y/dx^2 - 2dy/dx + y = x sin x

10. Explain the method of variation of parameters.

11. Solve x^2 y'' - xy' + y = 0 given that y_1 = x as a solution.

12. For a partial differential equation by eliminating the constants h and k from (x - h)^2 + (y - k)^2 + z^2 = c^2. (Out of Syllabus)

PART-B

(4 × 12 = 48 Marks — Answer all questions)

13. (a) Solve (2x + y - 3) dy = (x + 2y - 3) dx

(b) Solve dy/dx + x sin 2y = x^3 cos^2 y

14. (a) (i) It is found that 0.5 percent of radium disappears in 12 years. What percentage will disappear in 1000 years?

What is the half-life of radium?

(b) (i) Write short note on radio activity and carbon dating.

(ii) Find the family orthogonal to the family y = c e^(-x) of exponential curves. Determine the member of each family passing through (0, 4).

15. (a) Solve (D^2 - 2D + 1)y = x e^x sin x

(b) Solve (D^2 + 4D + 4)y = 3x e^(-2x) using the method of undetermined coefficients.

16. (a) Solve x^3 D^3 y + 3x^2 D^2 y + xDy + y = x + log x

(b) (1 + x^2) d^2y/dx^2 + (1 + x)dy/dx + y = 2 sin[log(1 + x)]

Osmania University — April/May 2025

Time: 3 Hours | Max. Marks: 80

PART-A

(8 × 4 = 32 Marks — Answer any eight questions)

1. Solve sqrt(1 + x^2) dx + sqrt(1 + y^2) dy = 0
2. Solve dy/dx = (x + 7y + 2)/(3x + 5y + 6)
3. Solve dx/x = dy/(x + z) = dz/(-z)
4. Solve xp^2 - 2yp + x = 0 where p = dy/dx
5. Find the singular solution of y = xp - p^2 where p = dy/dx
6. Find the orthogonal trajectories of x^2 + y^2 = a^2
7. Solve d^2y/dx^2 + 16y = 0
8. (3D^2 + D - 14)y = 13e^(3x) where D = d/dx
9. Find a particular integral of (D^2 + 4D + 3)y = cos 3x where D = d/dx
10. If y_1(x) = x is a solution of x^2 y'' + xy' - y = 0, find the general solution of y'(x) = x^2 y'' + xy' - y = x.
11. Find the partial differential equation by eliminating the arbitrary functions f, g from z = f(x + 2y) + g(2x - y). (Out of Syllabus)
12. Solve (x^2 D^2 + 3xD + 1)y = 0.

PART-B

(4 × 12 = 48 Marks — Answer all questions)

13. (a) Solve y dx + (x - y^3) dy = 0

(b) Solve (3x^2 y^4 + 2xy) dx + (2x^3 y^3 - x^2) dy = 0

14. (a) Solve y = 3px + 6p^2 y^2

(b) Solve y = yp^2 + 2px

15. (a) Solve (D^2 + 9)y = cos 3x + sin 2x

(b) Solve (D^2 + 4D + 4)y = e^(-2x) + x^2

16. (a) Using the method variation of parameter, solve (D^2 - 1)y = 1/(1 + e^x).

(b) Solve x^2 y'' - 6y = 5x^3 + 8x^2

Osmania University — June/July 2024

Time: 3 Hours | Max. Marks: 80

PART-A

(8 × 4 = 32 Marks — Answer any eight questions)

1. Solve x sqrt(1 + y^2) + y sqrt(1 + x^2) dy/dx = 0
2. Solve dy/dx = (x + 3y + 5)/(2x + 6y + 7)
3. Solve dx/(1 + x) = dy/(1 + y) = dz/z
4. Solve x^2 p^2 + 3xyp + 2y^2 = 0 where p = dy/dx
5. Find the general solution of y = xp + a/p
6. Find the orthogonal trajectories of y^2 = 4ax
7. Solve d^2y/dx^2 - 2dy/dx - 15y = 0
8. Solve (D^2 - 2mD + m^2)y = e^(mx) where D = d/dx
9. Find a particular integral of (D^2 - 3D + 2)y = sin 3x.
10. Solve 2x^2 y'' + 3xy' - y = 0 given that y_1 = 1/x is a solution.

11. Form a partial differential equation by eliminating the arbitrary constants a and b from z = ax^2 + by^2. (Out of Syllabus)

12. Solve (9x^2 D^2 + 3xD + 1)y = 0.

PART-B

(4 × 12 = 48 Marks — Answer all questions)

13. (a) Solve y(x + y) dx - x^2 dy = 0

(b) Solve (x^3 - 2y^2) dx + 2xy dy = 0

14. (a) Solve y - 2px = x^2 p^4

(b) Solve y = 2px + yp^2

15. (a) Solve (D^3 - 1)y = x e^x

(b) Solve (D^2 + 4)y = x cos x

16. (a) Using the method of variation of parameters, solve y'' + 9y = csc 3x.

(b) Solve x^3 y''' + 3x^2 y'' + xy' + 8y = 65 cos(log x)

Osmania University — June/July 2023

Time: 3 Hours | Max. Marks: 80

PART-A

(8 × 4 = 32 Marks — Answer any eight questions)

1. Solve y - x dy/dx = a(y^2 + dy/dx)

2. Solve (1 + x)dy/dx - xy = 1 - x

3. Solve dx/(1 + x) = dy/(1 + y) = dz/z

4. Solve (p + y + x)(xp + y + x)(p + 2x) = 0

5. Solve (y - px)(p - 1) = p

6. Solve y = yp^2 + 2px

7. Solve d^3y/dx^3 - 6d^2y/dx^2 + 11dy/dx - 6y = 0

8. Solve (D^2 - 4)y = x^2

9. Solve (D^2 - 3D + 2)y = 2x^2 + 3e^(2x) by using the method of undetermined coefficients.

10. Given that y = x is a solution of x^2 d^2y/dx^2 + x dy/dx - y = 0, x ≠ 0. Then find the general solution of x^2 d^2y/dx^2 + x dy/dx - y = x.

11. Explain Legendre's linear equation.

12. Form a partial differential equation by eliminating the constants h, k from (x - h)^2 + (y - k)^2 + z^2 = c^2. (Out of Syllabus)

PART-B

(4 × 12 = 48 Marks — Answer all questions)

13. (a) Solve (1 - x^2)dy/dx + 2xy = x sqrt(1 - x^2)

(b) Solve dx/[x(y^2 - z^2)] = dy/[-y(z^2 + x^2)] = dz/[z(x^2 + y^2)]

14. (a) Reduce xyp^2 - (x^2 + y^2 - 1)p + xy = 0 to Clairaut's form and find its singular solution.

(b) Bacteria in a certain culture increase at a rate proportional to the number present. If the number doubles in one hour, how long it takes for the number to triple?

15. (a) Solve (D^2 - 4D + 4)y = 8(x^2 + e^(2x) + sin 2x)

(b) Solve d^2y/dx^2 + 2dy/dx + y = x cos x

16. (a) Solve x^2 D^2 y - xDy - 3y = x^2 log x

(b) Solve (x^2 - y^2 - z^2)p + 2xyq = 2xz. (Out of Syllabus)

Osmania University — June/July 2022

Time: 3 Hours | Max. Marks: 80

PART-A

(8 × 4 = 32 Marks — Answer any eight questions)

1. Solve (1 - x)dy + (1 - y)dx = 0
2. Solve dy/dx - csc 2xy = 1/2 sec^2 x
3. Solve dy/(mz - ny) = dy/(nx - lz) = dz/(ly - mx)
4. Solve p^2 + 2py cot x = y^2
5. Solve y + px = x^4 p^2
6. Solve p = log(px - y)
7. Solve d^3y/dx^3 + 6d^2y/dx^2 + 11dy/dx + 6y = 0
8. Solve (D^2 + 2D + 1)y = 2x + x^2
9. Solve (D^2 + 2D + 4)y = 4x^2 + 6e^x by using method of undetermined coefficients.
10. Solve x^2 y'' - xy' + y = 0. Given that y_1 = x as a solution.
11. Explain Cauchy-Euler equation.
12. By eliminating the constants, find the partial differential equation from the relation 2z = x^2/a^2 + y^2/b^2. (Out of Syllabus)

PART-B

(4 × 12 = 48 Marks — Answer all the questions)

13. (a) Solve (1 + y^2) dx = (tan^(-1) y - x)dy

(b) Solve dx/[x(y^2 - z^2)] = dy/[y(z^2 - x^2)] = dz/[z(x^2 - y^2)]

14. (a) Solve x^2 p^2 + yp(2x + y) + y^2 = 0 by reducing it to Clairaut's form by using the substitution y = u and xy = v.

(b) Bacteria in a certain culture increase at a rate proportional to the number present. If the number N increases from 1000 to 2000 in 1 hour, how many are present at the end of 1.5 hours?

15. (a) Solve (D^2 - 2D + 5)y = e^(2x) sin x

(b) Solve (D^2 + 1)y = x^2 sin 2x

16. (a) Solve (D^2 + 4D + 4)y = 3x e^(-2x) by using the method of variation of parameters.

(b) Solve x^2 d^2y/dx^2 + x dy/dx - 4y = x^2 by using Cauchy-Euler equation.